

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – SUMMARY REPORT

COURSE NAME: ARTIFICIAL INTELLIGENCE **COURSE CODE:** CSA17

#	Problem	Method	Result
1	Water Jug	Breadth-First Search	3G filled first; 6 moves; 2 gal isolated in 4-gal jug
2	Mars Rover	PEAS + agent analysis	Model-Based Utility-Based Agent recommended
3	8-Queens	Backtracking Search	Columns 1-8 -> Rows 1,5,8,6,3,7,2,4 (valid)
4	OLA Cab Booking	Utility-based agent (scoring)	Prime selected, utility = 37.00
5	Logistics Network	Uniform Cost Search	S -> C -> D -> G, min cost = 9

1. Water Jug Problem

State represented as (4-gallon, 3-gallon). Starting from (0,0), BFS explored the reachable states level by level and returned the shortest sequence of fill/empty/pour moves. This run fills the 3-gallon jug first, reaching (2,0) in 6 moves — the 4-gallon jug ends with exactly 2 gallons.

2. Mars Rover

Answered as a PEAS-style analysis: percepts come from cameras, spectrometers and environmental sensors; the environment is partially observable, dynamic, sequential, continuous and single-agent; actions include driving, sampling and transmitting data; performance is measured by scientific yield, coverage, communication reliability, power efficiency and safety. A Model-Based Utility-Based Agent architecture was recommended since the rover must weigh competing options under partial observability.

3. 8-Queens Problem

Formulated as a state-space search (initial state, state space, actions, goal test, path cost) and solved with backtracking, which places one queen per column and abandons a partial placement the moment it becomes unsafe. The search returned the valid arrangement Column→Row = (1,1) (2,5) (3,8) (4,6) (5,3) (6,7) (7,2) (8,4).

4. OLA Cab Booking

Identified as a Utility-Based Problem-Solving Agent, since the booking decision must trade off comfort, fare and waiting time rather than just check reachability. Pseudocode queries nearby drivers per cab type, scores each by a weighted utility function, and books the highest-scoring option. In the worked example, Prime was selected (utility = 37.00) over the cheaper Micro/Mini/Shared options because of its short wait time and high comfort rating.

5. Logistics Network

Modeled as a weighted graph over warehouses S, A, B, C, D, G and solved with Uniform Cost Search, which always expands the lowest cumulative-cost frontier node via a priority queue. The search expanded S, A, B, C, D, G in that order and returned S→C→D→G at a minimum total cost of 9, which is cheaper than the more direct-looking S→A→D→G (cost 11) or S→B→G (cost 11).

All computational results above were produced by **U1_A1_Python_Code.py** and match the worked solutions in **U1_A1_Solution.pdf**.